National University

Of Computer & Emerging Sciences Faisalabad-Chiniot Campus

CS 1005 : Discrete Structures

Assignment 1

Instructor : Ms Mahzaib Younas

Section C

Q No 01: Find the negation of the proposition "Vandana's smartphone has at least 32 GB of memory" and express this in simple English and in expression.

Q No 2: Find the conjunction of the propositions p and q **(p ^ q),** where p is the proposition "Rebecca's PC has more than 16 GB free hard disk space" and q is the proposition "The processor in Rebecca's PC runs faster than 1 GHz."

Q No 03: What is the disjunction of the propositions p and q, where p is the proposition "Rebecca's PC has more than 16 GB free hard disk space" and q is the proposition "The processor in Rebecca's PC runs faster than 1 GHz."

Q No 04: Prove the logarithm Quotient law, if b=5, x=75 and y=125 respectively.

Q No 05: Describe the roles of addition with appropriate example?

Q No 06: Make a truth table for the statement

1. **$(P \lor Q) \land \sim(P \land Q)$**
2. **$(P \lor Q) \rightarrow (P \land Q)$**

Q No 07: Make a truth table and prove that given condition

$$\neg (P \lor (\neg P \land Q)) = (\neg P \land \neg Q)$$